

llmXive Automated Review of PhysisForcing: Physics Reinforced World Simulator for Robotic Manipulation

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided data and citations.

1. Ambiguity in Percentage Claims (Abstract, Introduction, Section 4.2): The paper repeatedly claims that PhysisForcing improves the Wan2.2-I2V-A14B model on R-Bench by “22.3%”.

- **Data Check:** Table 1 (R-Bench results) lists the base Wan2.2-I2V-A14B score as 50.7 and the PF-Wan score as 62.0.
- **Calculation:** The absolute difference is $62.0 - 50.7 = 11.3$. The relative improvement is $(11.3/50.7) \times 100 \approx 22.29\%$.
- **Issue:** While the math is correct for a *relative* increase, the phrasing “improving... by 22.3%” is ambiguous in scientific writing. It is often interpreted as an absolute percentage point gain (i.e., $50.7 + 22.3 = 73.0$). Given the context of benchmark scores, readers might misinterpret this as a massive absolute jump.
- **Recommendation:** The text should explicitly state “a 22.3% *relative* improvement” or “an 11.3 percentage point improvement” to ensure the claim is factually precise and not misleading.

2. “Best Overall Score” Claim (Abstract, Introduction, Section 4.2): The authors claim PF-Cosmos attains the “best overall score” on R-Bench, surpassing “all baselines including the strongest commercial model Wan2.6”.

- **Data Check:** Table 1 shows PF-Cosmos at 63.8. The highest score among non-PhysisForcing baselines is indeed Wan2.6 at 60.7.
- **Issue:** The claim is factually supported by the table *as presented*. However, the phrasing “surpassing all baselines” is a strong absolute claim. If the table omits any relevant state-of-the-art models (even if they are not the “strongest”), the claim could be challenged.
- **Recommendation:** While the claim is technically true based on the provided table, it is safer to phrase it as “surpassing all *evaluated* baselines” or “achieving the highest score among all reported models” to maintain strict factual accuracy regarding the scope of the comparison.

3. Precision of Margins (PAI-Bench and EZS-Bench): In Section 4.2, the paper states PF-Cosmos surpasses Abot-PhysWorld (84.9) with a score of 85.2 on PAI-Bench.

- **Data Check:** Appendix Table 2 (PAI-Bench) lists Abot-PhysWorld at 84.91 and PF-Cosmos at 85.17.
- **Issue:** The text rounds these to 84.9 and 85.2, implying a margin of 0.3. The actual margin is 0.26. While rounding is standard, the claim “surpassing... (84.9)” with a rounded “85.2” slightly exaggerates the visible gap.

- **Recommendation:** Ensure that when claiming a “surpass” or “best” status with very small margins (e.g., < 0.5 points), the text either provides the exact values or qualifies the statement to avoid implying a more significant lead than the data supports.

4. Citation Consistency: The paper cites [deng2026rethinking](#) for R-Bench and [ali2025abot](#) for EZS-Bench. The bibliography entries for these are present. The claims regarding the benchmark sizes (650 pairs for R-Bench, 196 for EZS-Bench) match the descriptions in the Appendix (Section 6). No factual discrepancies found in the citation-to-claim mapping for benchmark definitions.

Conclusion: The core scientific claims are supported by the data, but the presentation of percentage improvements and benchmark margins contains minor ambiguities that could lead to misinterpretation. These are writing-level issues affecting the precision of the claims rather than fundamental scientific errors.

Action items.

- **[writing]** Clarify the ‘22.3%’ improvement claim in the Abstract and Intro. Table 1 shows a relative increase (11.3/50.7), but the phrasing is ambiguous and could be misread as an absolute percentage point gain. Specify ‘relative improvement’ for precision.
- **[writing]** In Section 4.2, the claim that PF-Cosmos surpasses Abot-PhysWorld (84.9) with 85.2 implies a larger margin than the actual 0.26 point difference shown in Appendix Table 2. Use exact values or qualify the margin to avoid overstating the lead.
- **[writing]** The claim ‘surpassing all baselines’ in the Abstract is technically true for listed models but risks overgeneralization. Qualify as ‘surpassing all evaluated baselines’ to strictly align with the provided Table 1 data scope.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 effectively illustrates the method’s impact and presents clear benchmark results, but the caption’s definition of the method name for the WorldArena benchmark (‘PhysisForcing’) differs from the naming convention (‘PF-Cosmos’) used for the other benchmarks, which could cause confusion.

Figure 2

The figure provides a clear overview of the architecture and training flow, but the legend is incomplete; it defines ‘Video tokens’ but fails to define the specific icons used for low-level and high-level physics within the central block.

Figure 3

The bar chart effectively displays the average scores on PAI-Bench, but the x-axis labels are rotated at a steep angle that hinders readability. Additionally, the caption fails to define the ‘PF-Wan’ label shown on the axis, relying on definitions from Figure 1’s caption which are not explicitly cross-referenced here.

Figure 4

The figure presents a qualitative comparison of robotic video generation, but the visual results for the proposed method (PF-Cosmos) show a failure to execute the prompt, contradicting the caption’s claim of superior physical plausibility. Additionally, the baseline ‘Seedance 1.5 Pro’ row visually depicts a successful grasp, undermining the implied superiority of the proposed method.

Figure 5

The figure displays training curves with a legend, but the complete absence of a caption and undefined mathematical notation in the legend make the plot’s specific claims and components ambiguous.

Figure 6

The figure presents a grid of video frames comparing different models but lacks necessary labels and a descriptive caption to identify the specific methods or the task being performed.

Figure 7

Figure 7 presents a grid of robot manipulation frames but lacks any row labels or model identifiers in the image or caption, rendering the comparison meaningless without guessing from other figures. The generic caption fails to specify which models are shown or what the green border signifies.

Figure 8

The figure displays a grid of robot video frames but lacks row labels or a specific description in the caption to identify the models being compared or the task performed, rendering the comparison scientifically uninterpretable.

Figure 9

The figure displays a grid of robotic manipulation videos but lacks internal labels to identify the specific models in each row. The caption is too generic to stand alone, failing to describe the task or the specific baselines shown.

Figure 10

Figure 10 presents a grid of robotic manipulation results but fails to identify the compared models or describe the task, rendering the comparison scientifically meaningless despite the visual clarity of the individual frames.

Figure 11

Figure 11 effectively demonstrates the qualitative improvement of PF-Wan over the finetuned baseline across four robotic manipulation tasks. The visual comparison is clear, though the caption could be slightly more precise in mapping the ‘top’ and ‘bottom’ descriptions to the specific model labels shown in the images.

Figure 12

Figure 12 effectively demonstrates the qualitative improvement of the PF-Cosmos model over the finetuned baseline across four distinct robotic manipulation tasks. The layout is clear, with the baseline and proposed method clearly labeled and the proposed method highlighted in green as described in the caption.

Action items.

- **[writing]** Figure 1: The caption states that ‘WorldArena results use the Wan2.2-TI2V-5B world model trained with PhysisForcing,’ but the WorldArena chart in panel (c) labels the method as ‘PhysisForcing’ rather than ‘PF-Wan’ or ‘PF-Wan2.2’ to match the naming convention used for ‘PF-Cosmos’ in the other charts.
- **[writing]** Figure 1: The caption defines ‘PF-Cosmos’ for the R-Bench, PAI-Bench, and EZS-Bench charts, but does not explicitly define the label ‘PhysisForcing’ used in the WorldArena chart, creating a minor inconsistency in nomenclature.
- **[writing]** Figure 2: The legend in the top-right corner defines ‘Video tokens’ as a white square with a black border, but the diagram uses solid white squares (e.g., in the ‘PhysisForcing’ box) without borders, creating ambiguity between defined tokens and generic UI elements.
- **[writing]** Figure 2: The ‘PhysisForcing’ box contains icons for ‘Low-level physics’ (sine wave) and ‘High-level physics’ (brain) but lacks a corresponding legend entry to explicitly define these symbols.
- **[science]** Figure 3: The x-axis labels ‘PF-Wan’ and ‘PF-Cosmos’ are not defined in the caption, which only defines ‘PF-Cosmos’ as ‘Cosmos3-Nano trained with PhysisForcing’ in the context of Figure 1. The caption for Figure 3 fails to explain what ‘PF-Wan’ represents

or how the ‘PF-’ prefix relates to the baselines.

- **[writing]** Figure 3: The x-axis labels are rotated at a steep angle, making them difficult to read and cluttering the bottom of the chart. A horizontal layout or reduced font size would improve legibility.
- **[science]** Figure 4: The prompt specifies moving the apple onto the ‘second-level wooden platform,’ but the visual sequence for the proposed method (PF-Cosmos, bottom row) shows the robot failing to grasp or move the apple, leaving it on the table. This contradicts the caption’s claim that the method produces ‘more physically plausible’ interactions for this specific task.
- **[science]** Figure 4: The ‘Seedance 1.5 Pro’ row depicts the robot successfully grasping and lifting the apple, yet the caption implies this model fails to produce physically plausible interactions compared to the proposed method. The visual evidence in this row does not support the qualitative comparison made in the text.
- **[writing]** Figure 5: The figure has no caption provided (‘(no caption)’), making it impossible to verify what the plotted lines represent or what the specific metrics are without guessing.
- **[science]** Figure 5: The legend uses mathematical notation ($\mathcal{L}_{pix}^{phy}$, etc.) that is not defined in the figure or its missing caption, leaving the reader to guess the specific ablation components.
- **[science]** Figure 5: The plot shows a performance drop at 30k steps for all methods, but without error bars or a description of the experimental setup (e.g., number of seeds), it is unclear if this is a significant trend or noise.
- **[writing]** Figure 6: The caption ‘Comparison with the state-of-the-art models’ is generic and fails to specify the prompt, task, or specific models shown, unlike Figure 4 which details the prompt and method names.
- **[writing]** Figure 6: The image lacks row labels (e.g., model names) to identify which method corresponds to each row, requiring the reader to guess or cross-reference other figures.
- **[science]** Figure 7: The caption states ‘Comparison with the state-of-the-art models’ but does not identify which models correspond to the rows (e.g., Wan 2.6, Seedance 1.5 Pro, etc.) or define the green border around the bottom row, making the comparison impossible to interpret without external context.
- **[writing]** Figure 7: The caption is identical to those of Figures 6, 8, 9, and 10, providing no unique information to distinguish this specific qualitative comparison from others in the appendix.
- **[science]** Figure 8: The caption states ‘Comparison with the state-of-the-art models’ but does not identify which models correspond to the rows shown, unlike Figure 4 which explicitly lists them. Without this mapping, the comparison is uninterpretable.
- **[writing]** Figure 8: The caption is identical to Figures 6, 7, 9, and 10, providing no specific context or prompt for the visual content shown, making it impossible to assess if the results match the intended task.

- **[science]** Figure 9: The caption states this is a comparison with state-of-the-art models, but the image contains no text labels identifying which row corresponds to which model (e.g., Wan 2.6, Seedance 1.5 Pro, etc.), making it impossible to distinguish the baselines from the proposed method without external context.
- **[writing]** Figure 9: The caption is generic (‘Comparison with the state-of-the-art models’) and fails to describe the specific task shown (dual robotic arms capping a pen) or the specific models being compared, unlike Figure 4 which provides these details.
- **[science]** Figure 10: The caption states ‘Comparison with the state-of-the-art models’ but does not identify which specific models correspond to the rows shown, making the comparison uninterpretable.
- **[writing]** Figure 10: The image filename [Qual_Appx_6.jpg] suggests this is an appendix figure, yet the caption provides no context about what task or prompt is being demonstrated.
- **[writing]** Figure 11: The caption states ‘finetuned baseline (top)’, but the top row is labeled ‘Wan2.2-A14B (ft)’ while the bottom row is labeled ‘PF-Wan’. The caption should explicitly state that the top row is the finetuned baseline to match the visual labels.
- **[writing]** Figure 11: The caption describes the bottom row as ‘PF-Wan (bottom, green)’, but the green border is applied to the entire row of images, not just the text or a specific element. This phrasing is slightly ambiguous.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript demonstrates a strong command of the field but occasionally relies on specialized jargon that may hinder accessibility for a broader audience, particularly those outside the specific sub-fields of diffusion transformers and point tracking.

First, the acronym **DiT** (Diffusion Transformer) is used frequently (e.g., Abstract, Section 3.2, Section 3.3) without being explicitly defined at its first occurrence. While standard in recent literature, a brief expansion upon first use would significantly improve readability for non-specialists.

Second, the term “**physics-informative regions**” (Section 3.1) is used repeatedly. While descriptive, it is somewhat jargon-heavy. Phrases like “physically active regions” or “interaction-critical areas” might convey the same meaning more plainly. Similarly, the term “**backbone**” (Section 4.1, Appendix) is standard CV jargon for the base model architecture; replacing it with “base model” or “underlying architecture” would be more inclusive.

Third, standard deep learning acronyms like **MLP** (Section 3.2) and concepts like **flow matching** (Section 3.4) are used without definition. While these are fundamental to the authors’

domain, the paper’s goal of improving physical plausibility for a wider robotics community suggests that brief parenthetical definitions (e.g., “Multi-Layer Perceptron (MLP)”) would be beneficial.

Finally, the phrase “**vanilla finetuning**” (Section 4.1) is slightly informal jargon. “Standard finetuning” or “baseline finetuning” would be more precise and professional.

Addressing these points will ensure the paper’s contributions are accessible to the widest possible audience without diluting the technical precision.

Action items.

- **[writing]** Define ‘DiT’ (Diffusion Transformer) at its first occurrence in the Abstract or Introduction. Currently, it appears as ‘DiT features’ without prior definition, which excludes readers unfamiliar with this specific architecture acronym.
- **[writing]** Replace the term ‘physics-informative regions’ with a plainer alternative like ‘physically active regions’ or ‘interaction-critical areas’ in Section 3.1. The current phrasing is slightly redundant and could be simplified for broader accessibility.
- **[writing]** Define ‘MLP’ (Multi-Layer Perceptron) at its first use in Section 3.2. While common in deep learning, the paper aims for broad accessibility, and defining standard acronyms at first use is a best practice for non-specialist readers.
- **[writing]** Clarify the term ‘flow matching’ in Section 3.4. While ‘flow matching loss’ is used, a brief parenthetical explanation (e.g., ‘a generative training objective’) would help readers from non-generative-model backgrounds understand the context.
- **[writing]** Replace ‘backbone’ with ‘base model’ or ‘underlying architecture’ in Section 4.1. ‘Backbone’ is standard jargon in computer vision but may be opaque to general robotics or AI researchers not specialized in model architectures.

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The logical flow of the paper is generally sound, establishing a clear problem (physical implausibility in video generation) and proposing a hierarchical, region-focused solution. The connection between the proposed losses and the claimed improvements in benchmarks is supported by the ablation studies. However, there are two specific logical inconsistencies regarding the formulation of the method and the interpretation of the results that require clarification.

First, in Section 3.1 (“Physics-informative Region Extraction”), the authors define the foreground weight r_i in Equation 2 as $r_i = \frac{1}{D_0(p_i^0) + \epsilon}$. The text explicitly states that this weight is introduced because “robot-object interactions are usually distributed in foreground areas.” However, in standard depth maps, a larger D_0 value corresponds to objects further away

(background), while a smaller D_0 corresponds to objects closer to the camera (foreground). The formula $1/(D_0 + \epsilon)$ yields a *larger* value for *larger* D_0 (background) and a *smaller* value for *smaller* D_0 (foreground). This mathematical definition directly contradicts the textual claim that it prioritizes foreground regions. If the goal is to weight foreground (near) regions higher, the formula should likely be proportional to D_0 (if D_0 is distance from camera) or inverted in a way that prioritizes small depth values. As written, the mechanism selects background regions, which undermines the logical basis for the “region-focused” claim.

Second, in Section 4.2 (“Evaluation results on Policy Learning”), the authors claim that PhysisForcing improves downstream policy success, citing an average increase from 68.2% to 72.8%. While the average supports the claim, Table 1 reveals that the method significantly degrades performance on two specific tasks: “shake_bottle” (-3.0%) and “stack_bowls_two” (-6.5%). The text attributes the success to better handling of “contact-rich” tasks like placing and pressing. However, “stack_bowls_two” is inherently a contact-rich manipulation task involving precise physical interaction. The paper offers no logical explanation or hypothesis for why the physics alignment would fail or be detrimental in these specific contact-rich scenarios while succeeding in others. This omission leaves a gap in the causal argument that the method universally improves physical plausibility for manipulation; it suggests the method may introduce biases that are harmful in certain physical contexts, which is not addressed in the discussion.

Action items.

- **[science]** In Section 3.1 (Eq. 2), the foreground weight is defined as $r_i = 1/(D_0 + \epsilon)$. Since D_0 is a depth map where larger values indicate objects further away, this formula assigns higher weights to background (far) regions, contradicting the text’s claim that it weights ‘foreground areas’. The formula should likely be $D_0 / (D_0 + \epsilon)$ or similar to prioritize near objects.
- **[science]** In Section 4.2 (Policy Learning), the paper claims PhysisForcing improves downstream policy success. However, Table 1 (RoboTwin results) shows the method degrades performance on ‘shake_bottle’ (-3.0%) and ‘stack_bowls_two’ (-6.5%). The text attributes gains to ‘contact-rich’ tasks but fails to logically reconcile why the proposed physics alignment would harm performance on tasks that arguably also involve physical contact and manipulation, suggesting a missing analysis of failure modes.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper makes several claims that extrapolate beyond the immediate evidence provided in the benchmarks and ablation studies.

First, the claim that the method yields “stronger representations for robotic manipulation”

(Abstract, Conclusion) is not fully supported by the policy learning results. While the average success rate on RoboTwin 2.0 increases from 68.2% to 72.8% (Table 1), this improvement is not uniform; the method degrades performance on three of the six tasks (shake_bottle, adjust_bottle, stack_bowls_two). The paper attributes the gain to “stronger representations” but does not provide an ablation or analysis to distinguish whether the improvement stems from better physical understanding or simply higher visual fidelity that aids the specific downstream policy architecture (Fast-WAM). Without evidence that the feature space itself is more robust or generalizable, this causal link is an over-interpretation of the aggregate metric.

Second, the statement that PhysisForcing “surpasses all world-model planners” (Abstract, Sec 4.2) is an over-claim based on the provided data. In Table 2, while the average score of 24.0% is the highest, the Tesseract baseline achieves a significantly higher score on Task 2 (35.0% vs. 26.0%). By focusing solely on the average, the paper obscures the fact that the proposed method is not universally superior across all tasks. A more accurate claim would be that it achieves the highest *average* performance, rather than surpassing all planners in a general sense.

Finally, the assertion that the method “consistently improves” performance across backbones (Abstract) and “improves every backbone” (Sec 4.2) is slightly exaggerated when looking at the granular data in Table 1. For the Wan2.2-TI2V-5B backbone, the “Quadruped” embodiment score shows negligible improvement (59.0 -> 59.7) compared to the double-digit gains seen in other categories. While the average improves, the lack of consistent improvement across every single sub-category suggests the “consistently” qualifier should be tempered or qualified to reflect the variance in performance across different embodiments.

Action items.

- **[science]** The claim that PhysisForcing ‘yields stronger representations for robotic manipulation’ (Abstract, Conclusion) is over-extended. The policy improvement (Table 1) is marginal (+4.6% avg) and inconsistent (decreases on 3/6 tasks). The paper does not provide ablation evidence isolating whether the gain comes from ‘stronger representations’ or simply better video fidelity, nor does it rule out that the gains are specific to the Fast-WAM architecture rather than a general property of the video model.
- **[writing]** The assertion that the method ‘surpasses all world-model planners’ (Abstract, Sec 4.2) is an over-claim. Table 2 shows PhysisForcing (24.0%) beats WoW (20.5%) and Tesseract (18.0%), but Tesseract achieves 35.0% on Task 2. The paper frames the average as the sole metric of superiority without addressing the variance or the fact that Tesseract outperforms the proposed method on a specific task, suggesting the ‘surpassing’ claim is too absolute.
- **[writing]** The paper claims the method ‘consistently improves embodied video generation’ (Abstract) and ‘improves every backbone’ (Sec 4.2). However, Table 1 shows that for the Wan2.2-TI2V-5B backbone, the ‘Quadruped’ embodiment score drops from 59.0 (base) to 59.7 (ft) to 59.7 (PF), showing negligible or no improvement compared to the significant gains in other categories. The word ‘consistently’ implies uniform improvement across all dimensions, which the data does not fully support.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper addresses safety and ethics primarily through the “Broader impacts” section (Appendix A.6) and the nature of the training data. While the authors correctly identify the dual-use risk of generating highly realistic robotic manipulation videos (e.g., for fabricating deceptive footage or training policies on ungrounded synthetic data), the proposed mitigation is limited to a “research-only release.” Given the high fidelity of the generated videos, a more robust technical mitigation strategy should be outlined. Specifically, the authors should discuss the integration of invisible watermarking or provenance metadata (e.g., C2PA standards) into the generated outputs to prevent misuse in disinformation campaigns.

Regarding data ethics, the manuscript states the model is trained on a filtered subset of the “RoVid-X” dataset (Section 4.1, Appendix A.1). However, there is no explicit statement regarding the consent, licensing, or privacy compliance of the source videos. If RoVid-X contains footage of human operators, private workspaces, or proprietary industrial processes, the lack of transparency regarding data sourcing and potential IRB/ethics board oversight is a concern. The authors should clarify the provenance of the training data, confirm that all data is publicly available under appropriate licenses, and state whether any human subjects were involved in the data collection process.

The evaluation benchmarks (R-Bench, PAI-Bench, EZS-Bench) rely heavily on MLLM-as-Judge protocols (Appendix A.2). While not a direct safety failure, the reliance on automated judges for “physical plausibility” could inadvertently reward models that generate visually smooth but physically impossible scenarios if the judge model itself lacks rigorous physical grounding. The authors should briefly discuss the limitations of these automated safety/quality metrics in detecting subtle physical violations that could lead to unsafe policy learning.

Overall, the paper demonstrates awareness of ethical risks, but the mitigation strategies and data provenance details require expansion to meet rigorous safety standards for a model capable of generating high-fidelity robotic simulations.

Action items.

- **[writing]** The ‘Broader impacts’ section (Appendix A.6) acknowledges risks of deceptive footage and synthetic policy training but lacks concrete mitigation strategies beyond ‘research-only release.’ Explicitly detail technical safeguards (e.g., invisible watermarking, provenance metadata injection) or usage restrictions planned for the code release to address dual-use concerns.
- **[writing]** The training data (RoVid-X, 4M clips) is described as a ‘filtered subset’ but lacks details on consent, privacy, or ethical sourcing of the underlying robotic videos. Clarify if the dataset contains human operators or private environments and confirm compliance with data privacy standards (e.g., IRB approval or public license verification) to prevent potential

privacy violations.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The scientific evidence supporting the claims of PhysisForcing is generally strong, with consistent improvements across three distinct benchmarks (R-Bench, PAI-Bench, EZS-Bench) and two different backbone architectures (Wan2.2, Cosmos3). The ablation studies (Table 3) effectively demonstrate the complementary nature of the pixel-level and semantic-level losses, and the region-focused masking (Table 4) provides a plausible mechanism for the efficiency gains. The training dynamics (Figure 4) further support the stability of the proposed objective.

However, the validity of the “ground truth” used for supervision warrants closer scrutiny. The pixel-level alignment loss (Eq. 9) relies entirely on trajectories generated by CoTracker3 on the ground-truth videos. In complex robotic manipulation scenes involving occlusions, rapid motion, or textureless objects, CoTracker3 itself may produce erroneous tracks. If the model is trained to minimize the MSE against these noisy tracks, it risks learning to replicate tracker artifacts rather than true physical dynamics. The paper should include an analysis of the reference tracker’s error rate on the specific RoVid-X dataset or an ablation where the loss is computed against a different, perhaps physics-simulated, trajectory source to ensure the gains are not an artifact of the tracker’s biases.

Similarly, the semantic-level loss (Eq. 12) aligns features with a frozen V-JEPA 2 encoder. While V-JEPA is a strong self-supervised model, its training objective (predicting masked features) does not explicitly encode physical laws like conservation of momentum or contact rigidity. The claim that this alignment enforces “physical consistency” assumes that the encoder’s relational matrix implicitly captures these laws. The authors should provide evidence that the frozen encoder’s relational structure is indeed sensitive to physical interactions (e.g., by showing that the relational matrix changes significantly when a contact is broken vs. maintained) or include a control experiment where the alignment is performed with a non-physical visual encoder to isolate the contribution of the specific “physics” in the teacher model.

Finally, the policy learning results (Table 2) show a notable performance degradation on the `shake_bottle` and `stack_bowls_two` tasks (-3.0% and -6.5%). While the average improves, the lack of analysis for these specific failures is a gap in the evidence. It is unclear if the physics constraints are too rigid for these specific dynamics or if the region-masking fails to identify the relevant regions for these tasks. A brief discussion or sensitivity analysis regarding the hyperparameters for these specific tasks would strengthen the robustness claim.

Action items.

- **[science]** The pixel-level loss (Eq. 9) uses CoTracker3 trajectories as ground truth. Since CoTracker3 has errors on occlusions, the model may learn tracker artifacts. Quantify tracker

noise or ablate with a physics-simulated reference to prove gains reflect true physics.

- **[science]** The semantic loss (Eq. 12) aligns with V-JEPA 2, which is not explicitly trained on physical laws. Prove the encoder’s relational matrix encodes physical constraints (e.g., contact persistence) via qualitative examples or a control experiment with a non-physical encoder.
- **[science]** Policy results (Table 2) show performance drops on ‘shake_bottle’ (-3.0%) and ‘stack_bowls_two’ (-6.5%). Analyze why physics alignment degrades these specific tasks and discuss trade-offs or hyperparameter sensitivity to ensure robustness across the full task distribution.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical analysis presented in the paper is generally sound in its design but lacks necessary rigor in reporting uncertainty and significance testing, which is critical for claims of “consistent improvements” across multiple benchmarks.

1. Lack of Uncertainty Quantification: The primary results in Table 1 (R-Bench), Table 2 (PAI-Bench), and Table 3 (EVS-Bench) report single-point average scores (e.g., PF-Cosmos: 63.8, 85.2, 81.1). Given that R-Bench evaluates 650 prompts and the policy experiments involve 200 rollouts per task, the variance across these samples is non-trivial. The absence of standard deviations, standard errors, or 95% confidence intervals makes it impossible to determine if the observed gains (e.g., +9.2% over base) are statistically significant or within the noise of the evaluation metric. For instance, in Table 4 (RoboTwin), the task `shake_bottle` shows a decrease of 3.0% and `stack_bowls_two` a decrease of 6.5%. Without variance estimates, we cannot assess if the average gain of +4.6% is robust or driven by a few outliers.

2. Missing Significance Testing: The paper claims “consistent gains” and “surpassing strong baselines” but does not report the results of any statistical hypothesis tests (e.g., paired t-tests, Wilcoxon signed-rank tests) comparing PhysisForcing against the baselines. In the policy learning section (Sec 4.2), where the sample size is explicitly stated (200 rollouts), a formal test of significance is required to support the claim that the improvement is real and not due to random chance. The current presentation treats the benchmark scores as deterministic values rather than estimates of a distribution.

3. Sensitivity of Region Extraction: In Section 3.1, the physics-informative mask is generated using an adaptive threshold based on the *mean* score of the physics-informative metric (Eq. 3). The choice of the mean is arbitrary; if the distribution of motion scores is skewed (e.g., a few very fast-moving objects), the mean may not represent the “typical” interaction region well. The paper should either provide a justification for using the mean or include a sensitivity

analysis showing that the final performance is robust to different thresholding strategies (e.g., median, top-k percentile).

Recommendation: The authors should re-run the evaluation to compute and report standard deviations for all benchmark scores. Additionally, they should perform statistical significance tests for the key comparisons (PhysisForcing vs. Baseline) and report p-values. Finally, a brief sensitivity analysis on the mask thresholding method would strengthen the methodological rigor.

Action items.

- **[science]** Report uncertainty metrics (e.g., standard deviation or 95% confidence intervals) for all benchmark scores in Tables 1, 2, and 3. The current presentation of single-point averages (e.g., 63.8, 85.2) obscures the variance across the 650 R-Bench prompts or 200 policy rollouts, making it impossible to assess statistical significance of the reported gains.
- **[science]** Clarify the statistical test used to validate the improvements in Table 4 (RoboTwin) and Table 5 (WorldArena). Specifically, state whether the reported success rate differences (e.g., +4.6% average) are statistically significant ($p < 0.05$) given the sample size (200 rollouts/task) and the observed variance in the baseline performance.
- **[science]** In Section 3.1 (Eq. 3), the adaptive threshold for the physics mask is defined as the mean score. Justify this choice statistically or provide an ablation on the sensitivity of the final results to this threshold (e.g., using median or top-k percentile) to ensure the region selection is robust and not an artifact of the specific thresholding method.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a high standard of academic writing, with clear, concise, and logically structured prose. The narrative flow from the problem statement in the Introduction to the proposed method and experimental validation is coherent and easy to follow. Technical terms are generally well-defined, and the distinction between pixel-level and semantic-level alignment is articulated effectively throughout the text.

However, a few minor issues regarding clarity and notation precision were identified that, while not obstructing understanding, could be refined for a polished final version. In the Introduction, specifically in the fourth paragraph, the reference to “these signals” in the sentence regarding uniform supervision is slightly ambiguous; it would benefit from a more explicit antecedent to ensure the reader immediately grasps that the “signals” refer to the localized physical supervision cues. Additionally, in Section 3.1, the mathematical notation for rounding (Eq. 4) utilizes a non-standard bracket pair ($\lfloor \cdot \rceil$). While the text defines this, standardizing the notation to a more conventional form (e.g., $\lfloor \cdot \rfloor$) would prevent potential confusion for a broader audience. Finally, the caption for Table 1 in the Experiments section

relies on color coding to indicate rankings. To ensure full accessibility and clarity, the caption should explicitly state that the colored backgrounds correspond to the 1st, 2nd, and 3rd best results, ensuring the information is conveyed even if the visual distinction is lost.

Overall, the writing quality is strong, and these points represent minor refinements rather than substantive flaws. The paper is well-suited for its intended audience with these small adjustments.

Action items.

- **[writing]** In the Introduction (Section 1, paragraph 4), the sentence ‘Applying supervision uniformly over all pixels dilutes these signals and weakens alignment’ is slightly ambiguous. Clarify whether ‘these signals’ refers to the physical supervision signals or the physical evidence mentioned in the previous sentence to improve logical flow.
- **[writing]** In Section 3.1 (Eq. 4), the notation uses $\lfloor \cdot \rfloor$ for rounding. While defined in the text, the symbol $\lceil \cdot \rceil$ is non-standard for rounding (usually $\lfloor \cdot \rfloor$ or $\lceil \cdot \rceil$ is not a standard pair). Consider using standard notation or explicitly defining the rounding operator to avoid confusion for readers unfamiliar with this specific LaTeX macro.
- **[writing]** In Section 4 (Table 1 caption), the phrase ‘Per-column rankings are highlighted in...’ is followed by color box commands. Ensure the text explicitly states that the colors correspond to 1st, 2nd, and 3rd place to ensure accessibility for readers who may not see the colors or if the PDF is printed in grayscale.